

Figure 1: Two Centroids

The results for two centroids seem to be a bit blurry, however some distinct characteristics are there. It is reasonable to say that since there's a lower number of centroids it wasn't able to capture a lot of the groupings of the different digits, but it still managed to capture some groupings. For example, centroid 2 seems to be distinctly a digit 9 (though you can see some traces of a seven in there), while centroid one is less distinct and is an amalgamation of eight, zero, and six. Compared to eigenvectors these centroids create a much more recognizable digit. However, that wouldn't be a fair comparison because eigenvectors are designed to capture the direction of variations and then use that to reconstruct the image.

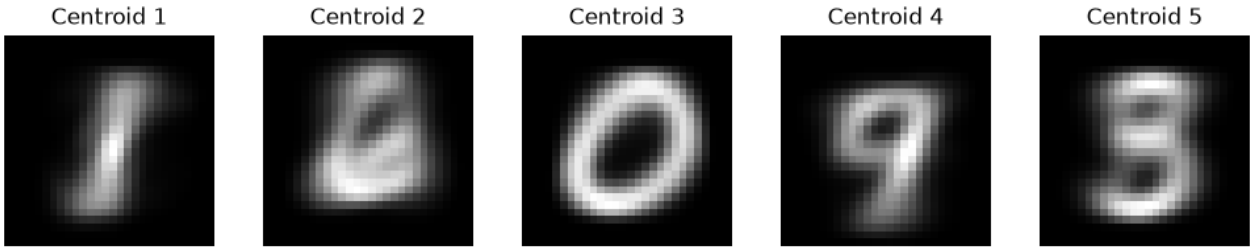


Figure 2: Five Centroids

The results for five centroids are much more distinct than the results for two centroids. Because there are more centroids this allows them to capture a lot more of the pixel groupings. This is why centroid one is distinctively the digit one, three is zero, and four is nine. Centroid two is an amalgamation of six and some other digits (I believe zero?), while centroid five is a mix between three and seven. Since there are more centroids this means that wild fluctuations are reduced because an outlier that is far away can be attributed to another centroid. One thing that I noticed was that these images look remarkably similar to reconstructed images that only used five to twenty eigenvectors.

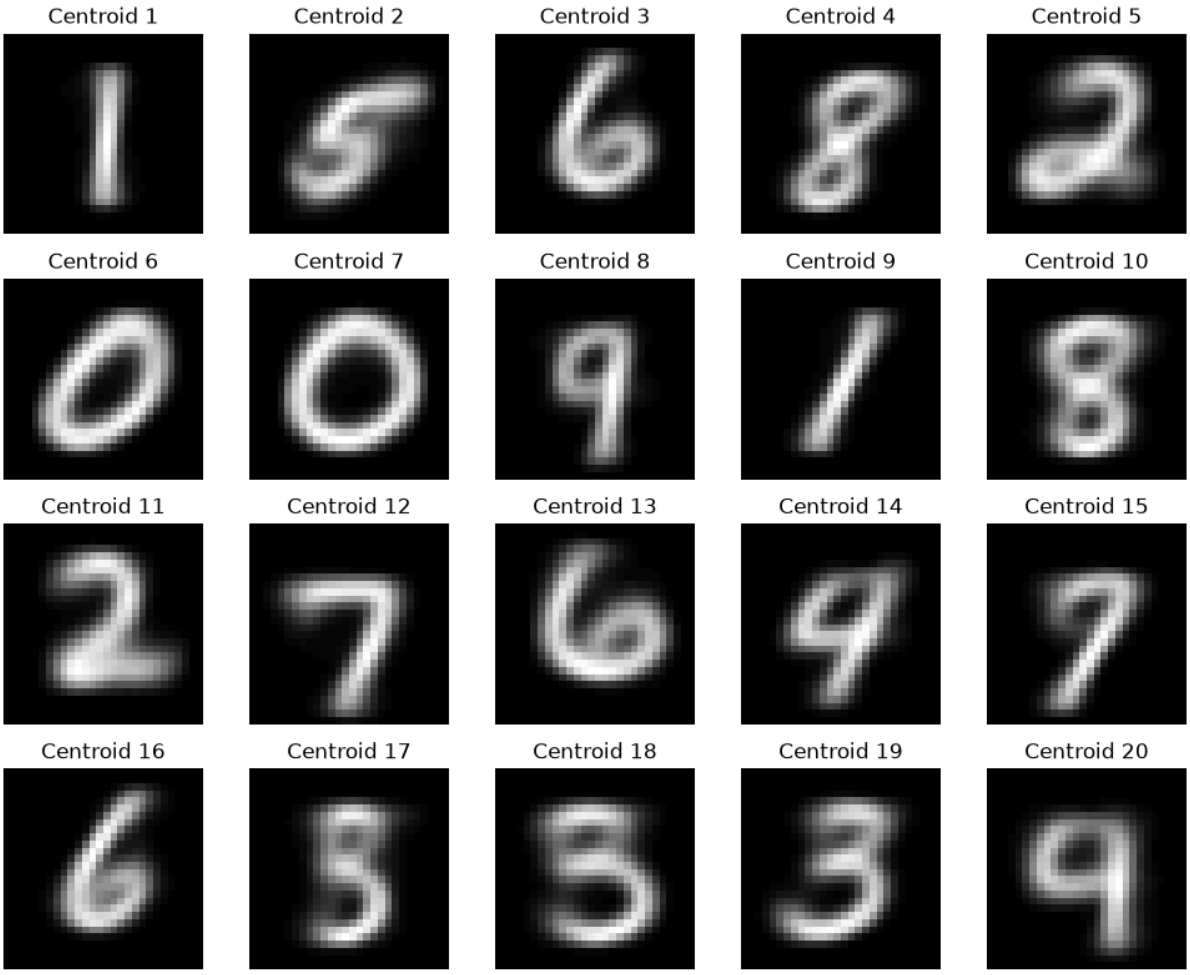
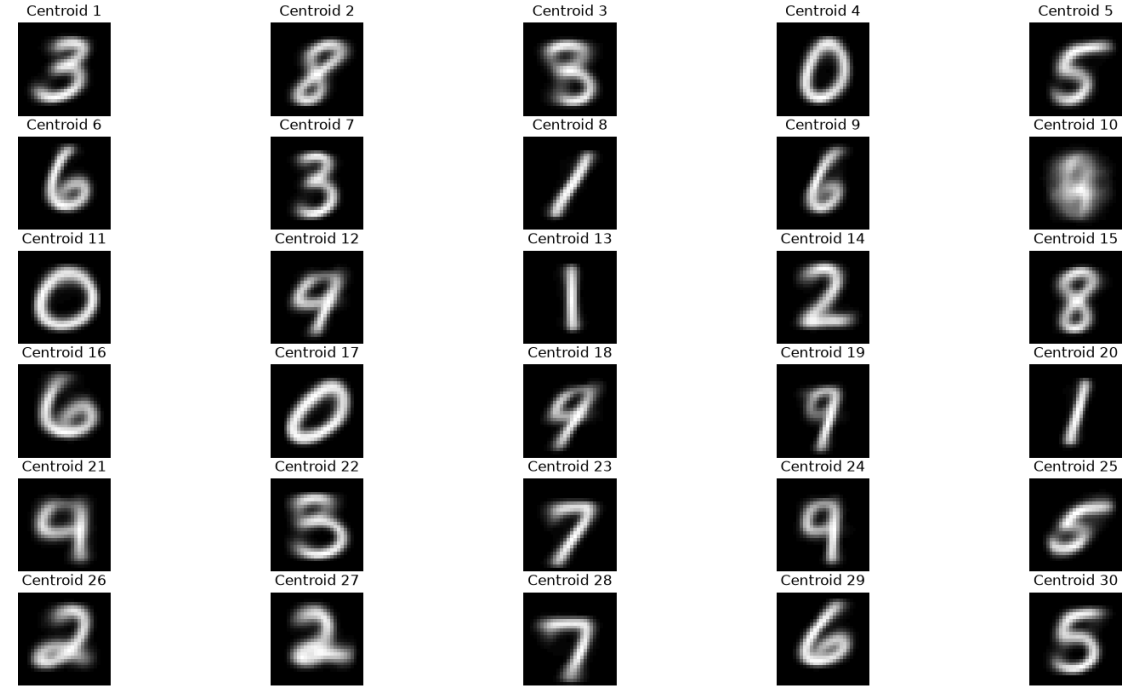


Figure 3: Twenty Centroids

Since there are more available centroids than digits we can start to notice a pattern. The way that K-means work is by first randomly assigning centroids, and then taking the sum squared distances between the points and the centroids. Afterwards it moves the centroids to reduce the distance. So overall centroids capture groupings of pixels, and because these are handwritten digits there are going to be different distinct ways to write the same number. This causes what we see with centroid seventeen, eighteen, and nineteen with all three of them being different ways to write the number three.


Centroid 1
Centroid 2
Centroid 3
Centroid 4
Centroid 5
Centroid 6
Centroid 7
Centroid 8
Centroid 9
Centroid 10
Centroid 11
Centroid 12
Centroid 13
Centroid 14
Centroid 15
Centroid 16
Centroid 17
Centroid 18
Centroid 19
Centroid 20
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Centroid 22
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Centroid 28
Centroid 29
Centroid 30

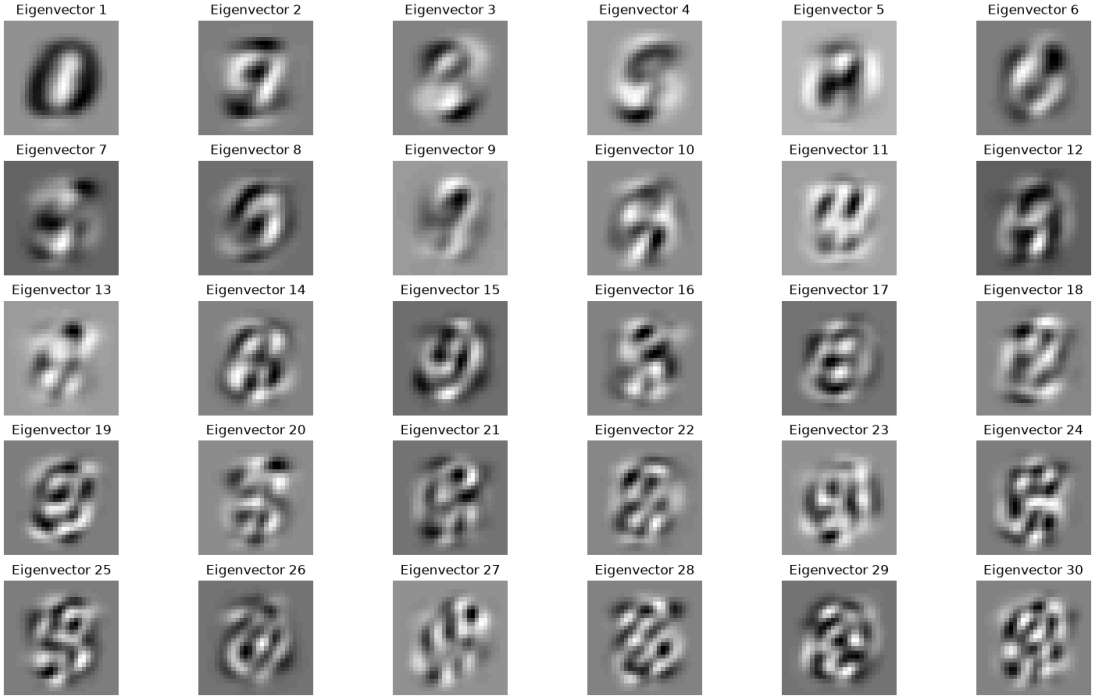

Eigenvector 1
Eigenvector 2
Eigenvector 3
Eigenvector 4
Eigenvector 5
Eigenvector 6
Eigenvector 7
Eigenvector 8
Eigenvector 9
Eigenvector 10
Eigenvector 11
Eigenvector 12
Eigenvector 13
Eigenvector 14
Eigenvector 15
Eigenvector 16
Eigenvector 17
Eigenvector 18
Eigenvector 19
Eigenvector 20
Eigenvector 21
Eigenvector 22
Eigenvector 23
Eigenvector 24
Eigenvector 25
Eigenvector 26
Eigenvector 27
Eigenvector 28
Eigenvector 29
Eigenvector 30

Figure 4: Thirty Centroids and Thirty Eigenvectors

With thirty centroids we notice the same exact thing. There are multiple centroids that represent the same digit, but different ways of writing it. However, the more interesting point is centroid 10. We went over this in class, where if we were able to view all images as a dot on a dot graph, we could pick centroids that would be the “most optimal.” However, we aren’t able to do that because of the massive amount of data, and because of this situations like centroid ten happen. K-means randomly assigns where the centroids should be and then tries to minimize the sum squared distance between centroids. So situations arise where a centroid will have a less distinct grouping than others. In this case, centroid ten seems to look more like an eigenvector than a centroid. As discussed in previous analysis, it isn’t really a one-to-one comparison between eigenvectors and centroids as they are tracking different things. Eigenvectors are trying to find and capture the direction of variations, while centroids are trying to group together pixels. Because of this centroids produce more recognizable digits than eigenvectors. While the eigenvectors can be used together to produce reconstructed digits. Another interesting thing to note is that, increasing the number of centroids allows for capturing more distinct groupings of pixels. However, increasing the number of eigenvectors allows us to capture more variations and use that to reconstruct even more accurate images.

AI Acknowledgement

Specific Tools: ChatGPT and Visual Studio Code built in GitHub Copilot

ChatGPT: I used ChatGPT as a way to estimate how long the homework will take as well as help me understand some concepts of the homework. It also answered any questions I had about K-means and helped debug and make my code more efficient.
https://chatgpt.com/share/6ad10d26-3ad8-83ea-82f5-970f04a3b860

GitHub Copilot: Visual Studio Code has a built-in GitHub Copilot agent that helped with endline completion of code and found any mistakes I made while coding.
https://github.com/AkkU438/K_means